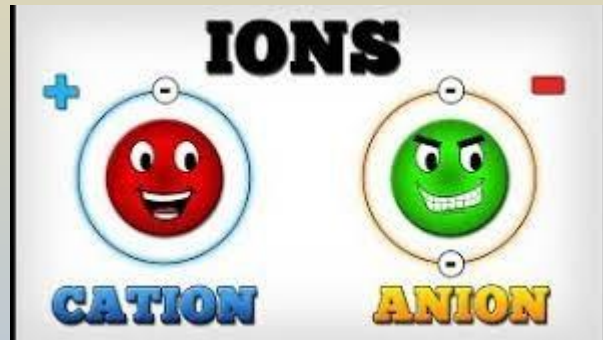


Pharmaceutical Analytical Chemistry I

Course Code (PA 102 & CPA 102)



Anion

Cation

Lecture (3)

**Carbonate & Bicarbonate
Group
Sulphur Containing Anions
Group**

Dr. Maha Mahmoud

**Lecturer in Pharmaceutical Analytical
Chemistry Department, MTI University**

Learning Outcomes (LOs)
By the End of This Lecture ,
Students will be able to know:

**1- Carbonate & Bicarbonate Group
Sulphur Containing Anions Group**

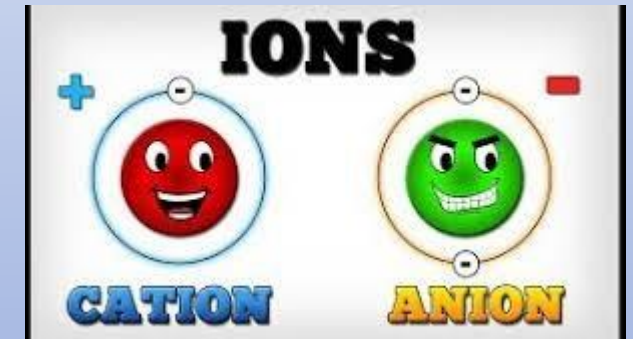
(Members, Symbol & Charge)

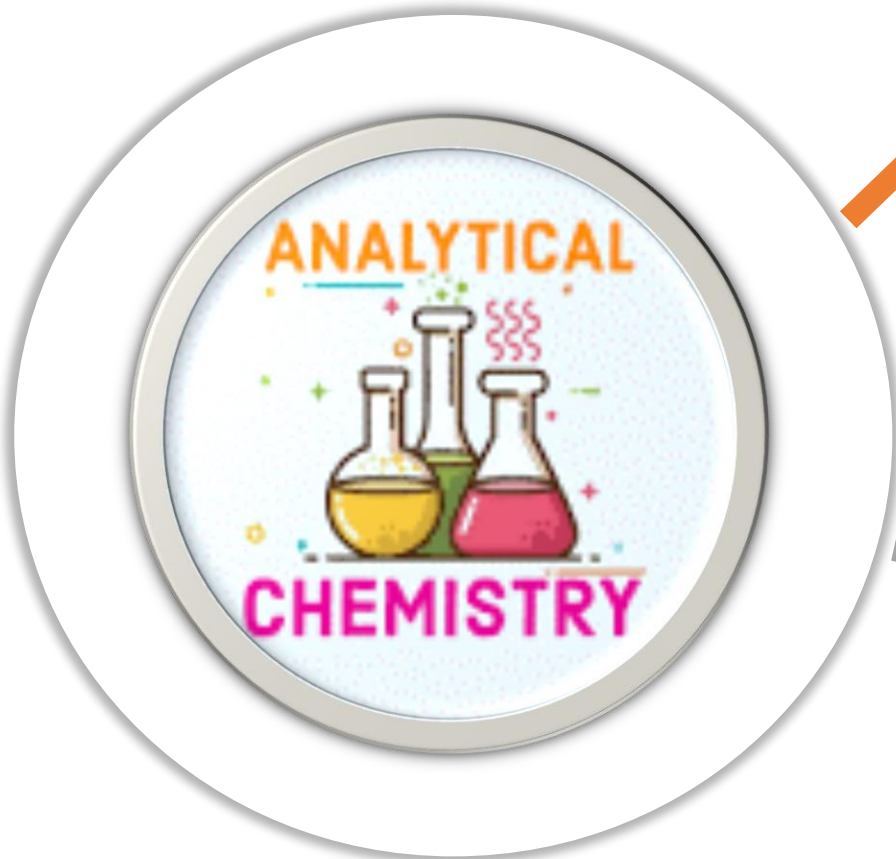
2- Dry Reaction of Both Groups

3- Wet Reaction of Both Groups

4- Confirmatory Test for Both Groups

5- How to Analyze Mixture





Qualitative Analysis

Identification of unknown through

- Physical Examination
- Chemical Examination



Quantitative Analysis

Determination of Concentration

Anions are Classified into 6 Groups



Carbonate & Bicarbonate Group

Carbonate & Bicarbonate

1

Sulphur Containing Anions Group

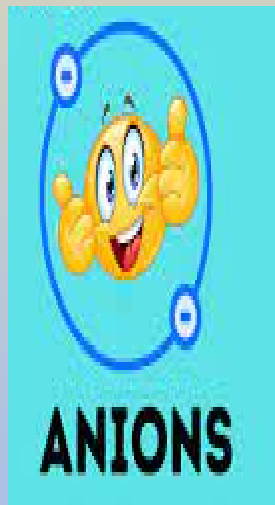
Sulphide, Sulphite,
Thiosulphate & Sulphate

2

Halides or Halogens

Fluoride, Chloride, Bromide & Iodide

3



4

Cyanogen Containing Anions Group

Cyanide, Thiocyanate,
Ferrocyanide & Ferricyanide

5

Phosphorus & Arsenic Containing Anions Group

Phosphate, Arsenate & Arsenite

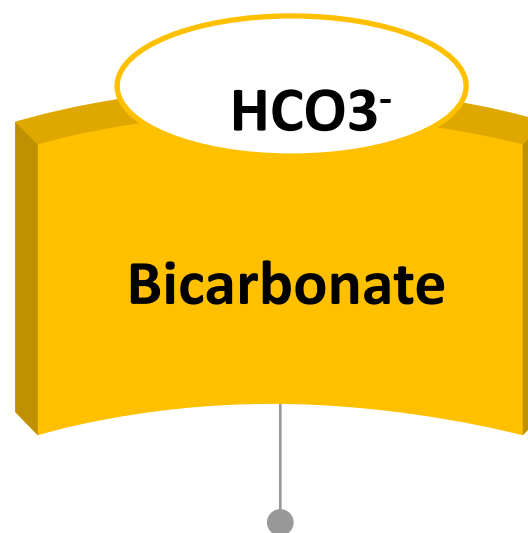
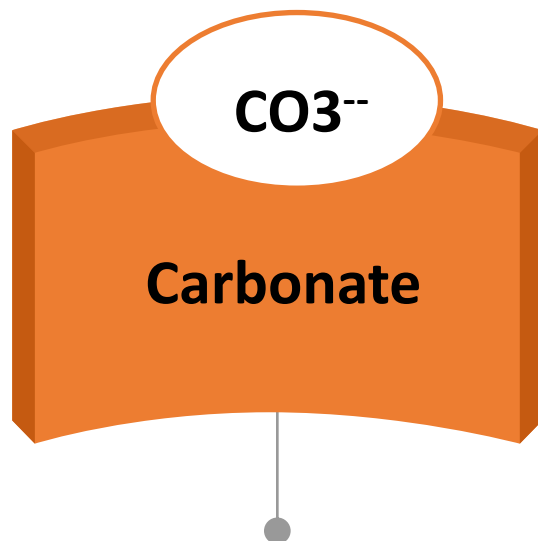
6

Nitrogen Containing Anions Group

Nitrate & Nitrite

1- Carbonates & Bicarbonates

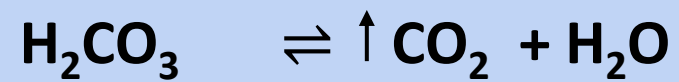
Carbonate & Bicarbonate Anions



The reaction of carbonates & bicarbonates are mostly similar.

Parent Acids

Carbonic acid
(H₂CO₃)
Weak Unstable Acid



A)- Dry Reaction

In dry reaction, the solid sample is treated with dilute hydrochloric and/or dilute sulphuric acid.

1- Action of dil. HCl



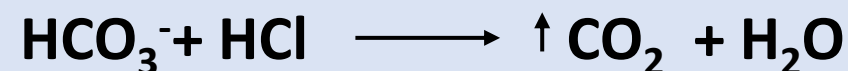
1)- Action of Diluted Hydrochloric Acid

Dil. HCl

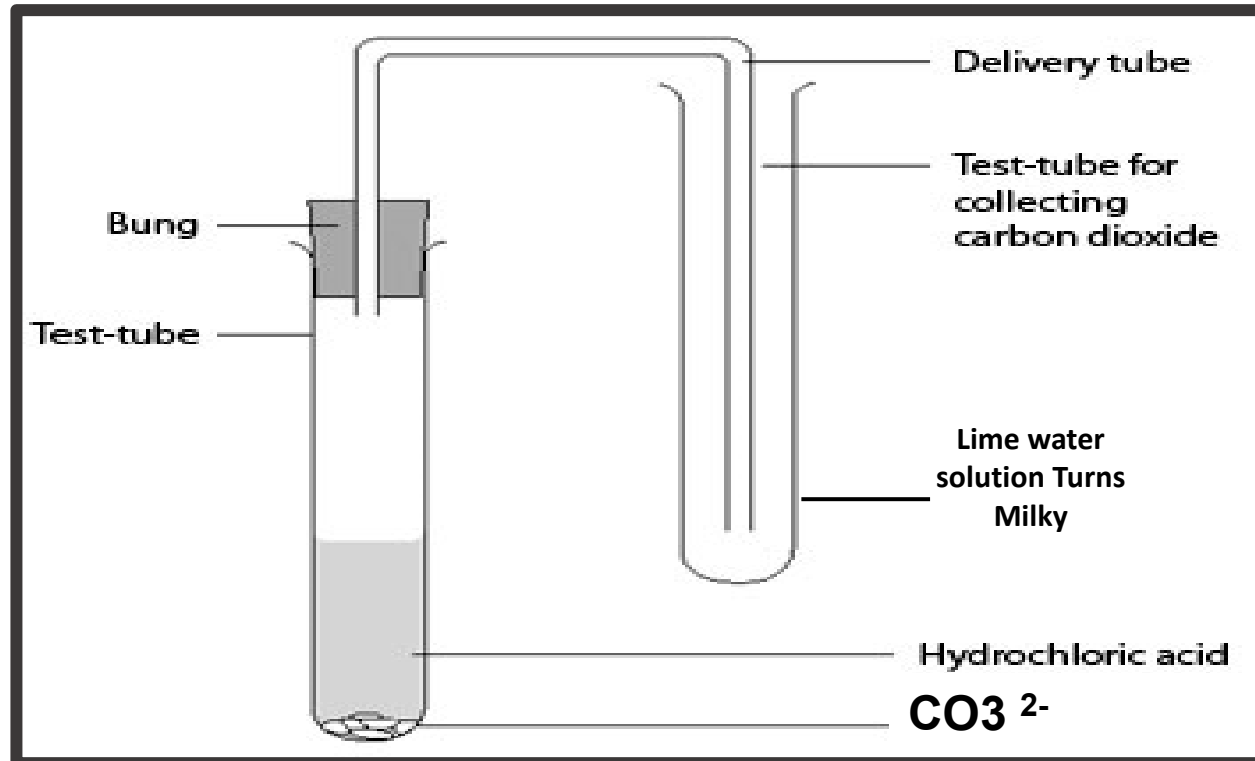
A)- CO_3^{--}

B)- HCO_3^-

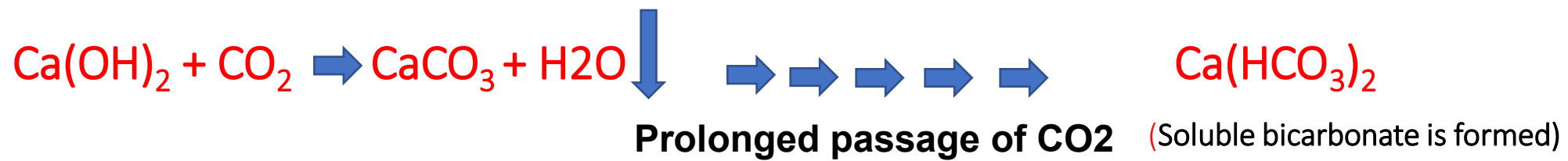
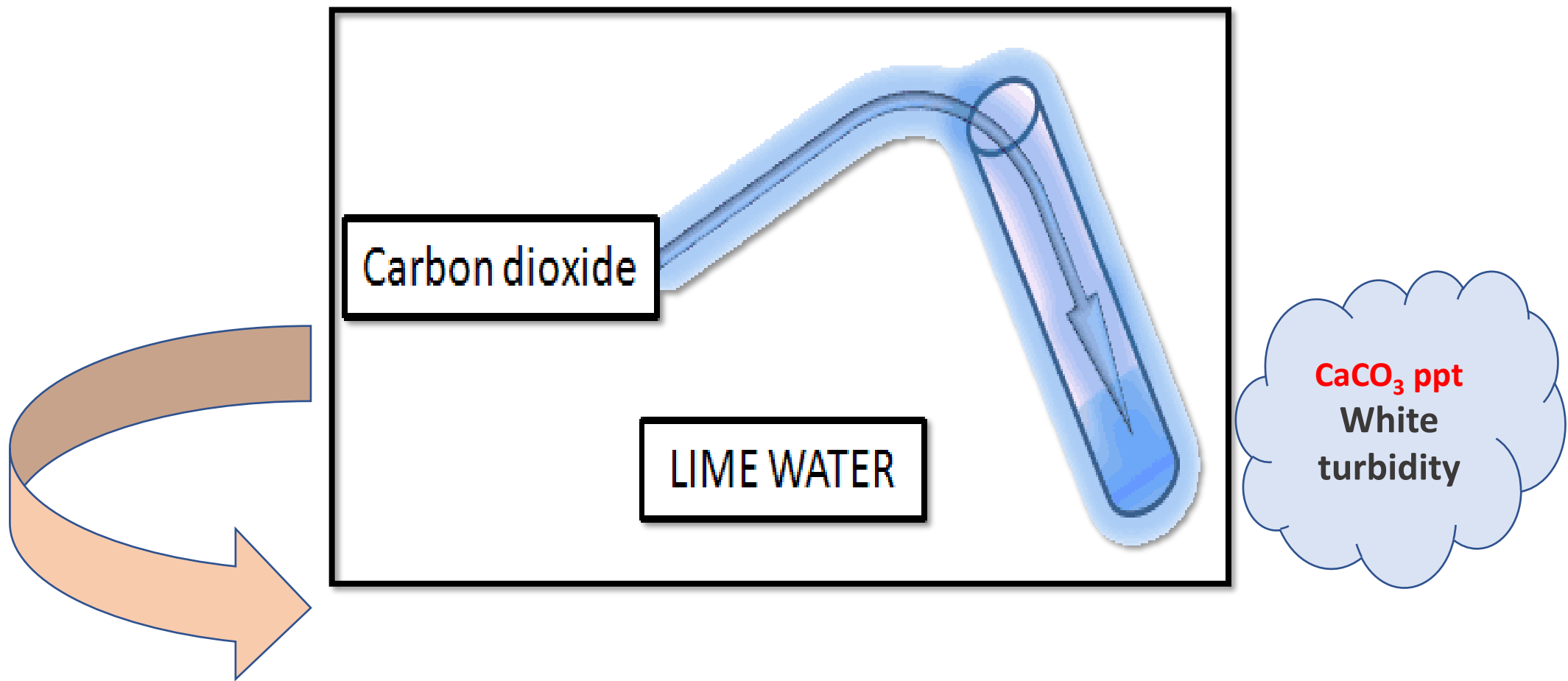
Decomposition with effervescence due to the evolution of CO_2 gas for both CO_3^{--} & HCO_3^-



Turbidity Test for Carbon Dioxide (Lime water test)



This test is considered to be selective for CO₂ and SO₂ gases.



B)-Wet Reaction

In order to carry out the wet reactions, a solution of the substance in water must be done (S.S)



Wet Reaction



**1- Effect of
Silver Nitrate**

**2- Effect of
Barium Chloride**

Solubility

All normal carbonates with the exception of those of the alkali metals & of ammonium are insoluble in water.

All bicarbonates are soluble in water.

1)- Reaction with Silver Nitrate

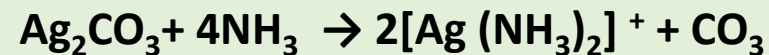


- White precipitate of silver carbonate is formed



The precipitate is soluble
in nitric acid & in ammonia

Give Reason: Ppt. soluble in excess Ammonia ??????



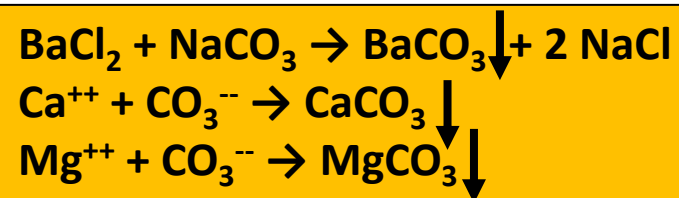
The precipitate becomes yellow or brown upon addition of excess reagent owing to the formation of silver oxide.



2)- Reaction with Barium Chloride

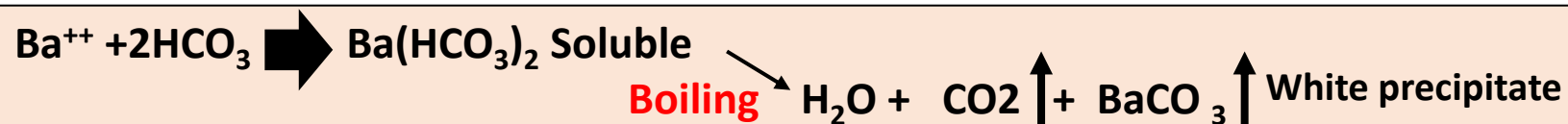
White precipitates of BaCO_3 , CaCO_3 and MgCO_3 will be obtained upon the addition of these reagents to samples of carbonate solution.

CO_3^{2-}



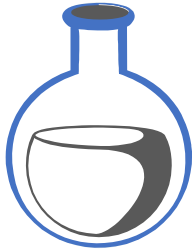
HCO_3^-

For HCO_3^- (No ppt. on cold since all bicarbonates are soluble in water)



Mixture of CO_3^{--} & HCO_3^-

1



Add excess CaCl_2 , BaCl_2 or MgSO_4 to
a solution of the mixture
 $\text{CO}_3^{--} / \text{HCO}_3^-$
(white ppt. indicates CO_3^{--})
Make Centrifuge



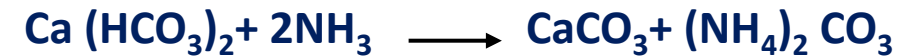
ppt. (CO_3^{--})
White ppt. CaCO_3
+
 H^+
↓
 $\text{CO}_2 + \text{H}_2\text{O}$
(Carbonate)



Centrifugate or Filtrate (HCO_3^-)

1) Boil
or

2) Treatment with ammonia Solution → white
ppt. is formed (Bicarbonate)



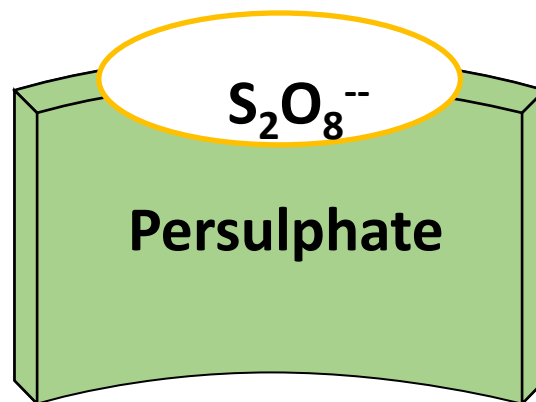
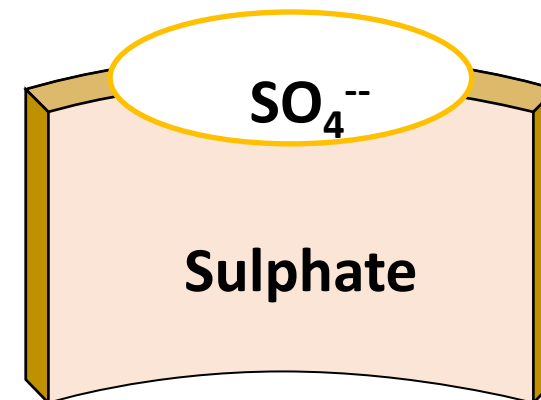
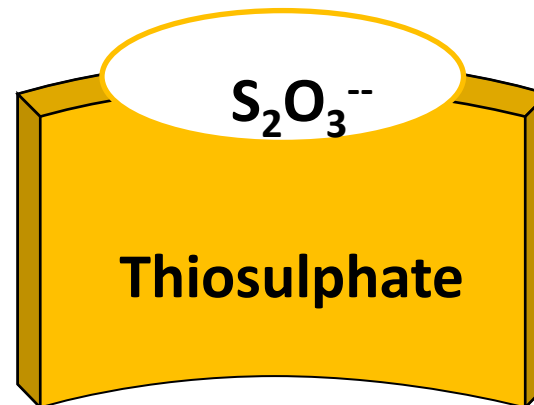
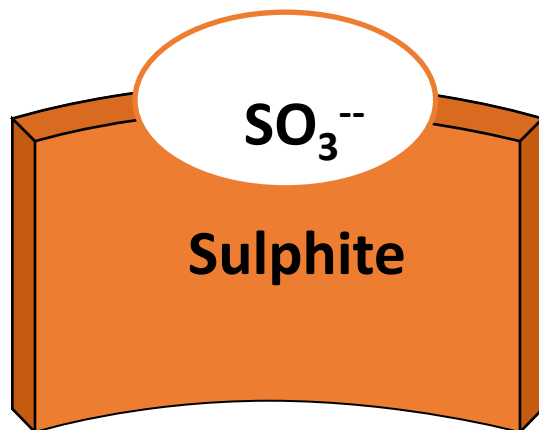
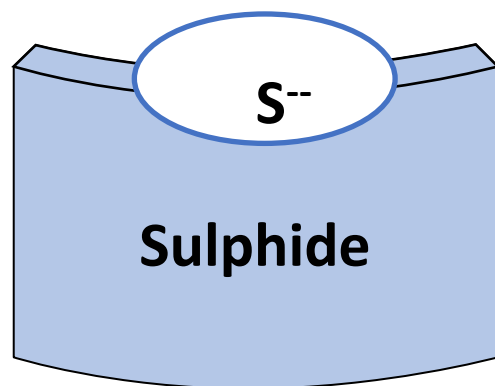
Summary



Test	CO_3^{-2}	HCO_3^-
Dry Reaction		
Solid + dil. HCl	Effervescence with evolution of colorless gas CO_2 , which causes turbidity of lime water, disappears after prolonged passage.	
Wet Reactions		
1- S.S + AgNO_3 Reagent	White ppt. of Ag_2CO_3	No ppt on cold
2- S.S + BaCl_2 , MgSO_4 & CaCl_2	All these reagents form white ppt. of the corresponding metal carbonates soluble In mineral acids.	No ppt. on cold. While after boiling white ppt. is formed

2- Sulphur Containing Anions

Sulphur Containing Anions



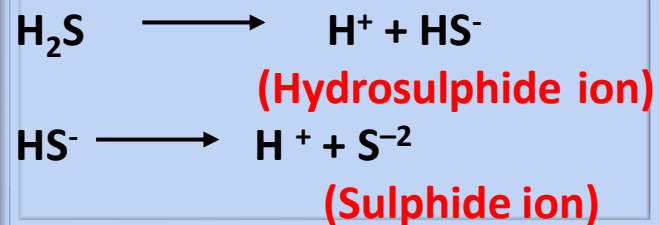
Parent Acids

Hydrogen Sulphide or
Hydrosulphuric Acid



1

- It is a gas with rotten egg odor.
- Ionized in two steps:



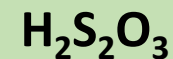
Sulphurous acid



2

- Diprotic acid.
- Moderate strong acidity

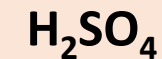
Thiosulphuric acid



3

- Stronger than sulphurous acid.

Thiosulphuric acid



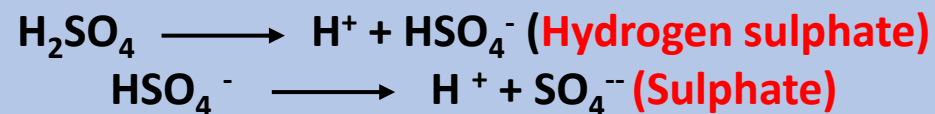
4

- Colourless oily liquid.
- Very strong acid.

General Properties of H₂SO₄

1- Acid properties

- It is one of the strongest acids, ionized in two steps:



2- Dehydrating properties

- Conc. H₂SO₄ has a great tendency to combine with water to form stable hydrates H₂SO₄ x H₂O. So, it is used as a dehydrating agent for certain substance & used mostly in the desiccators.

3- Oxidizing properties

- Considered as strong oxidizing agent .
- When heated with most reducing agents, It is reduced to SO₂



A)- Dry Reaction

1- Action of dil. HCl



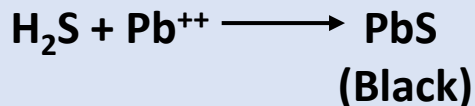
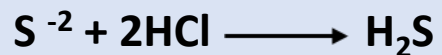
1)- Action of Diluted Hydrochloric Acid

A)- S^{-2}

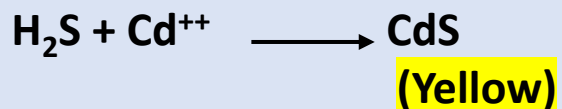
- Evolution of H_2S gas
- The gas evolved has rotten egg odor.

H_2S can be identified by:

a) - Blackening filter paper moistened with lead acetate solution.



b)- With filter paper moistened with cadmium acetate solution, turns yellow.



B)- SO_3^{-2}

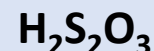
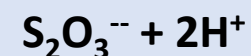
- Evolution of SO_2 .
- SO_2 gas has a burnt Sulphur odor.



- SO_2 turbid lime water (like CO_2)

C)- $S_2O_3^{-2}$

- No immediate change on cold.
- On warming with dil. HCl



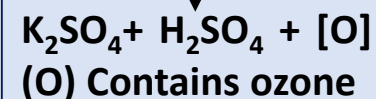
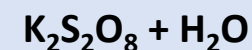
Yellow Colloidal

D)- SO_4^{-2}

No reaction with dil. HCl

E) $S_2O_8^{-2}$

- No effect on cold.
- On boiling with water, $S_2O_8^{--}$ decomposes to give oxygen which oxidize HCl.



Reducing Properties of S^{2-} & SO_3^{2-} & $S_2O_3^{2-}$

1- S^{2-}

It reacts with I_2 solution, acid $KMnO_4$, acid $K_2Cr_2O_7$ & behaves as reducing agent. It bleaches the brown color of I_2 solution, changes the pink color of acid $KMnO_4$ into colorless and changes the orange color of acid $K_2Cr_2O_7$ into green.



2- SO_3^{2-}

SO_2 like H_2S , bleaches the brown color of iodine, reacts with acid $KMnO_4$ and acid $K_2Cr_2O_7$.



3- $S_2O_3^{2-}$

-The Same

Solubility

All Na^+ , K^+ and NH_4^+ salts of sulphur containing anions are soluble in water

Sulphides

Other sulphides are in-soluble except those of Ca^{++} , Ba^{++} , & Sr^{2+} dissolve due to hydrolysis

Sulphites

Other sulphites are all in-soluble

Thiosulphates

Most $\text{S}_2\text{O}_3^{--}$ are soluble, Ag^+ , Hg^{2+} & Ba^{++} salts are slightly soluble.

Sulphates

All sulphates are soluble except pb^{++} , Ba^{++} , Sr^{++} , Ca^{++} & Mg^{++} salts are slightly soluble

Persulphates

All known persulphates are soluble.

B)-Wet Reaction

In order to carry out the wet reactions, a solution of the substance in water must be done



Wet Reaction



**1- Effect of Barium
Chloride**

**2- Effect of Silver
Nitrate**

**1- Effect of Ferric
Chloride**

**2- Effect of Lead
Acetate**

1)- Reaction with Barium Chloride



No Visible Reaction



White ppt. of $BaSO_3$ is formed which is soluble in dil. HCl.
 $Ba^{++} + SO_3^{-2} \longrightarrow BaSO_3 \downarrow$



No ppt. in dilute solution, but a ppt. is formed from very concentrated solution.



A white ppt. of $BaSO_4$ is formed which is insoluble in dil. HCl, even upon boiling.

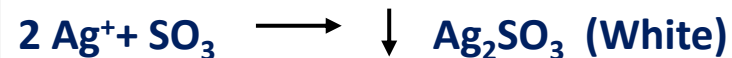
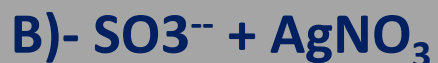
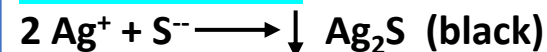


No ppt. on cold. On boiling a white ppt. of $BaSO_4$ due to the decomposition of $S_2O_8^{--}$ with the production of SO_4^{-2}
 $S_2O_8^{-2} + H_2O \xrightarrow{\text{boiling}} SO_4^{-2} + H_2SO_4 + (O)$

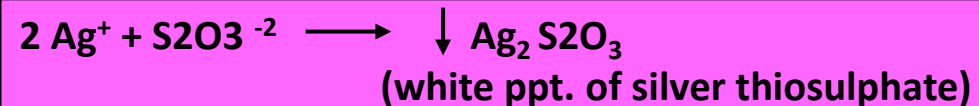
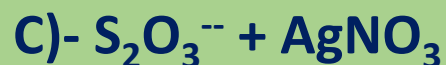
2)- Reaction with Silver Nitrate



On cold, black ppt. of Ag_2S is formed which is soluble in hot dil. HNO_3 , insoluble in ammonia & KCN solutions



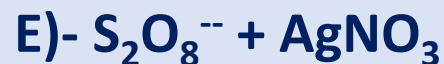
Silver sulphite is soluble in nitric acid, ammonia & in excess sulphite to give a complex salt.



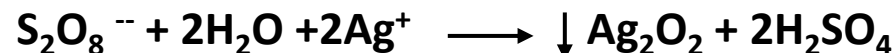
On standing white ppt. changes to yellow, brown and finally black, due to the formation of Ag_2S .



No ppt. in dil solution, but a ppt. may be formed in a very concentrated solution.



On cold no reaction with $AgNO_3$. On warming, a black ppt. of silver peroxide is formed



White

Yellow

Orange

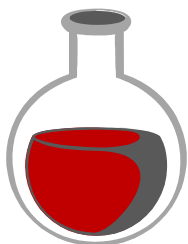
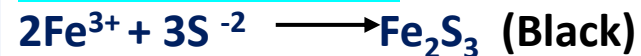
Brown

Black ppt. of Ag_2S

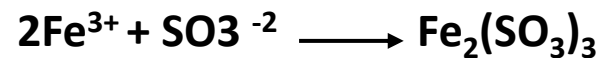
3)- Reaction with Ferric Chloride



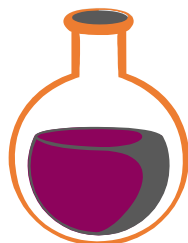
A black ppt. of Fe_2S_3 (**Ferric Sulphide**) is formed which is soluble in dil. Acids



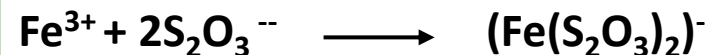
A **dark red colour** of (**Ferric sulphite**) is produced on cold.



Upon boiling a **reddish brown ppt**, of the basic iron salt is formed.



A **purple colour** of (**Complex Ferric Thiosulphate**) is produced



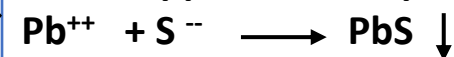
SO_4^{--} & $S_2O_8^{--}$: Do not react with $FeCl_3$.

4)- Reaction with Lead Acetate

A)- S^{--} + Lead Acetate



A Black ppt. of **PbS** is produced



A)- SO_3^{--} + Lead Acetate



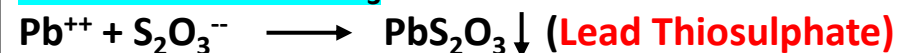
A white ppt. of **Lead sulphite** which is **soluble in cold HNO_3** .



A)- $S_2O_3^{--}$ + Lead Acetate



A white ppt. of **Lead thiosulphate** is formed which is **soluble in cold HNO_3**



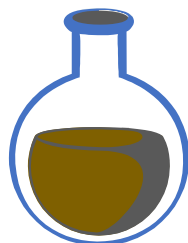
A)- SO_4^{--} + Lead Acetate



A white ppt. **Lead sulphate**, which is **insoluble in cold dil. mineral acids, but soluble in ammonium acetate and hydroxide solutions (Na^+ and K^+)**

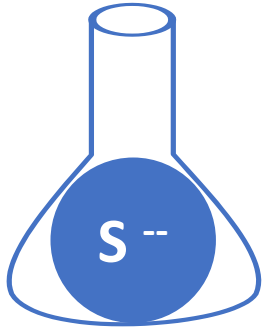


A)- $S_2O_8^{--}$ + Lead Acetate



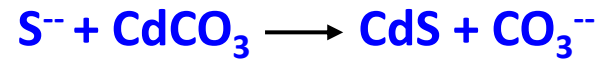
No reaction on cold, on boiling a **brown ppt.** of the hydrated peroxide appears.

Special Tests



Cadmium Carbonate Test

The sulphide solution is shaken with CdCO_3 , a **Canary Yellow ppt.** of CdS is produced.



Zinc Nitroprusside Test

Saturated ZnSO_4 Soln.

+

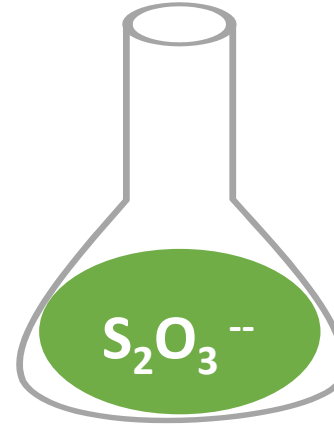
equal volume of



+

few drops of 1% sodium nitroprusside solution.

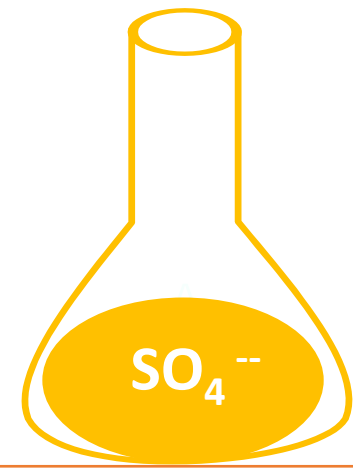
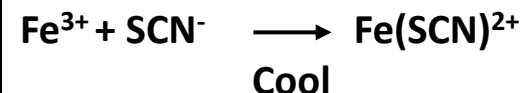
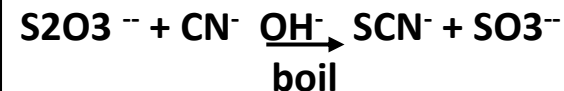
Then the solution is added to the SO_3^{--} salmon-coloured ppt. of zinc nitroprusside is formed $\text{Zn}(\text{Fe}(\text{CN})_5\text{NO})$.



Formation of Thiocyanate

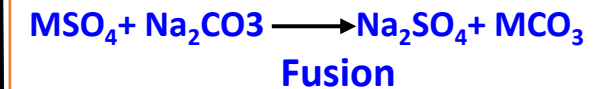
By boiling with KCN solution (poison), in the presence of NaOH , Cool, acidify & add FeCl_3

a blood red colour of ferric thiocyanate complex is produced.

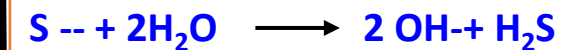


Hepar's test

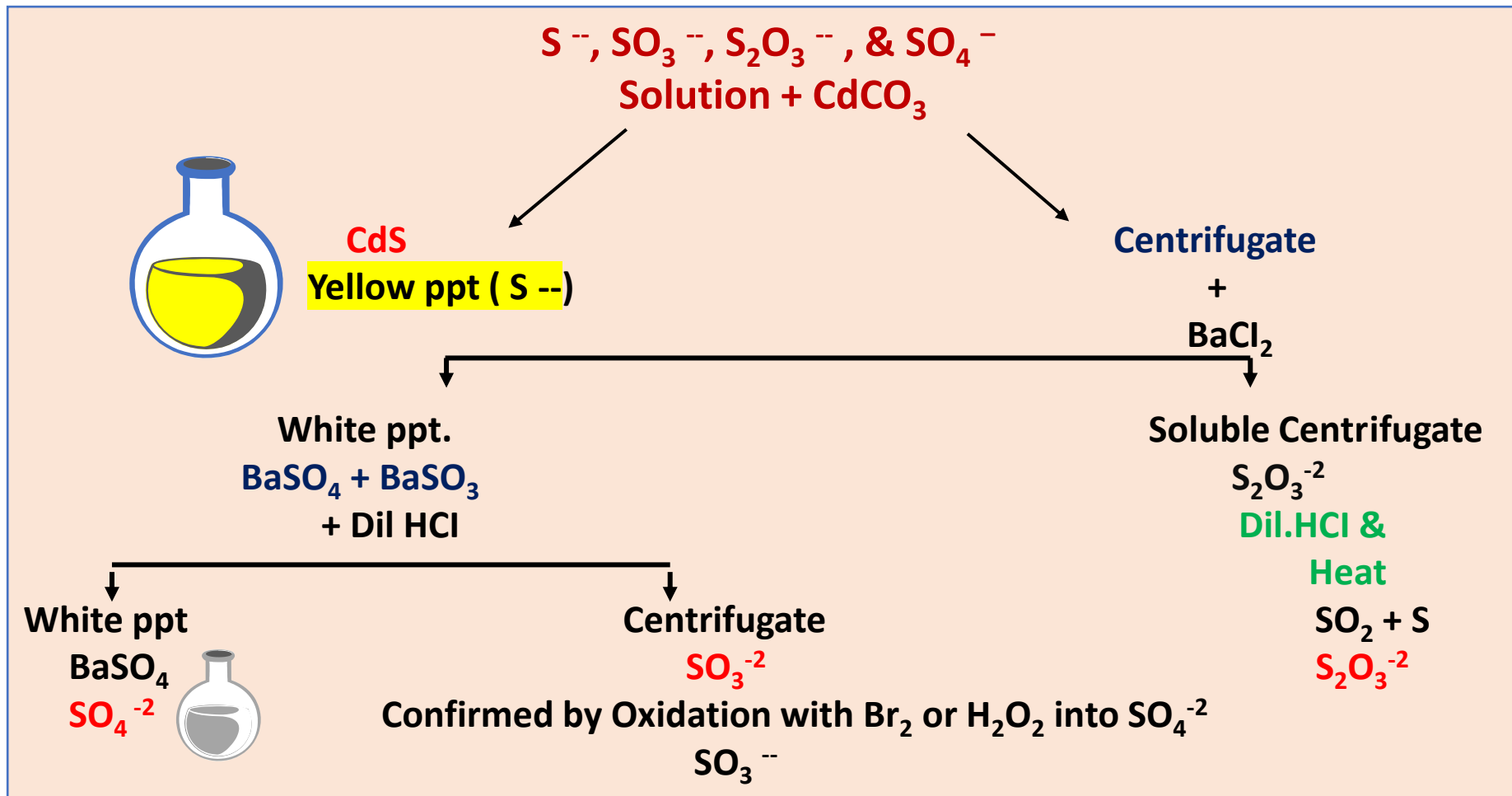
Sulphate is reduced by carbon to sulphide by heating on a piece of charcoal in the presence of Na_2CO_3 in the reducing zone of the flame.



$\text{Na}_2\text{SO}_4 + \text{C} \longrightarrow \text{Na}_2\text{S} + 4\text{CO}$
Transfer the fusion product to a silver coin and moisten with a little water, a **brownish black stain** of Ag_2S results.



1- Mixture of S^{--} , SO_3^{--} , $S_2O_3^{--}$ & SO_4^{--}



2)- Mixture of H₂S and SO₂ gases



In order to differentiate between these two gases which, evolve upon the addition of dil. HCl to sulphides, sulphites and thiosulphates and having similar reducing properties.

- A paper moistened with lead acetate solution changes into black when exposed to H₂S gas,
- SO₂ can cause turbidity to lime water.

3- Mixture of CO_3^{--} & SO_3^{--} or $\text{S}_2\text{O}_3^{--}$

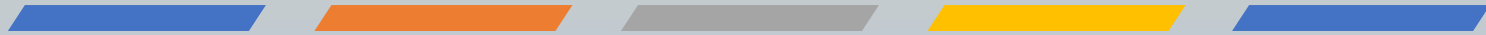
* This type of mixtures are considered to be difficult, due to the interference occur upon the addition of dil.

HCl which liberates CO_2 and SO_2 gases which turbides lime water and disappears on prolonged passage.

SO_2 can be detected by its reducing characters as discussed before, but CO_2 has non reduced characters.

Therefore SO_3^{--} or $\text{S}_2\text{O}_3^{--}$ ions must be firstly oxidized into SO_4^{--} by an oxidizing agent such as H_2O_2 , $\text{K}_2\text{Cr}_2\text{O}_7$ or KMnO_4 and dil. H_2SO_4 and warm, CO_2 will only evolve which can be test with lime water.

Summary



Test	S--	SO ₃ --	S2O3--	SO4--	S2O8--
Dry Reactions					
1- Solid + dil. HCl Cold & hot	Small Effervescence with evol. Of H ₂ S gas, rotten egg odour & has reducing properties	Burnt sulphur Odour of SO ₂ , Which has reducing Properties and turbides lime water	On warming Standing, SO ₂ gas evolves & the solution becomes turbid (colloidal S)	No reaction	On boiling Evolution of O ₂ , Cl ₂ which turn starch/kl Into blue.
Wet Reactions					
1- AgNO3 Reagent	Black ppt. of Ag ₂ S, sol. In hot dil. HNO ₃ , insol. In ammonia and KCN.	White ppt. of Ag ₂ SO ₃ , sol. In dil. Ammonia and excess SO ₂ on boiling black ppt. of Ag ₂ O	White ppt. of Ag ₂ S ₂ O ₃ sol. In excess S ₂ O ₃ ⁻² , colour change from yellow → black	No ppt.	On warming conc. solution Form black ppt. of Ag ₂ O ₂
2- BaCl₂ Reagent	No ppt	White ppt. of BaSO ₃ , sol. In dil. HCl, on standing or Warming with H ₂ O ₂ or Br ₂ → BaSO ₄	No ppt.	White ppt. of BaSO ₄ , insol. in Mineral acids, even on Boiling	On boiling a white ppt. of BaSO ₄ Is formed insol. In dil HCl
3- FeCl₃ Reagent	Black ppt. of Fe ₂ S ₃ , sol. In dil. acids	Dark red colour of (Fe ₂ (SO ₃) ₃), On boiling → reddish Brown ppt.	Purple colour of (Fe(S ₂ O ₃) ₂)- which Disappears on boiling	No Reaction	No Reaction
4- Lead Acetate Reagent	Black ppt. of Pbs, sol. In dil, HNO ₃	White ppt. of PbS ₂ O ₃ sol. In cold HNO ₃ On boiling gives → PbSO ₄	White ppt. of PbS ₂ O ₃ sol, in cold HNO ₃ on. Boiling → black ppt. of PbS	White ppt. Insoluble PbSO ₄ only sol. In am. Acetate or NaOH	On boiling Brown ppt. of Hydrated peroxide

